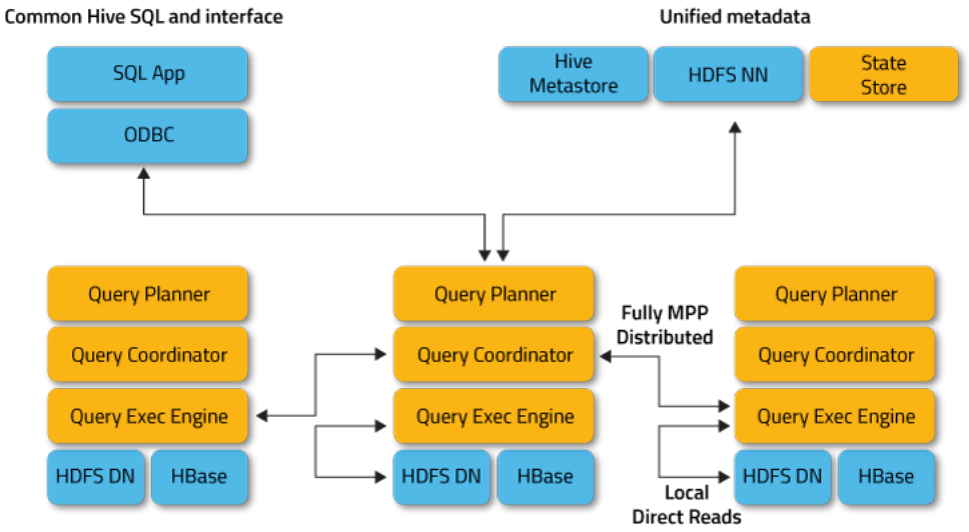
Apache Impala

Overview

Impala raises the bar for SQL query performance on Apache Hadoop while retaining a familiar user experience. With Impala, you can query data, whether stored in HDFS or Apache HBase – including SELECT, JOIN, and aggregate functions – in real time. Furthermore, Impala uses the same metadata, SQL syntax (Hive SQL), ODBC driver, and user interface (Hue Beeswax) as Apache Hive, providing a familiar and unified platform for batch-oriented or real-time queries. (For that reason, Hive users can utilize Impala with little setup overhead.)

Architecture

To avoid latency, Impala circumvents MapReduce to directly access the data through a specialized distributed query engine that is very similar to those found in commercial parallel RDBMSs. The result is order-of-magnitude faster performance than Hive, depending on the type of query and configuration.



There are many advantages to this approach over alternative approaches for querying Hadoop data, including::

- Thanks to local processing on data nodes, network bottlenecks are avoided.
- A single, open, and unified metadata store can be utilized.
- Costly data format conversion is unnecessary and thus no overhead is incurred.
- All data is immediately query-able, with no delays for ETL.
- All hardware is utilized for Impala queries as well as for MapReduce.
- Only a single machine pool is needed to scale.

We encourage you to read "Impala: A Modern, Open-Source SQL Engine for Hadoop" (http://cidrdb.org/cidr2015/Papers/CIDR15_Paper28.pdf) for details about Impala's architecture.

COMMUNITY CODE (http://www.apache.org/events/current-event.html)

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